

1 Schema Induction



Schema Induction

Induce Dimensions



Event



Location



Date



Damage



Impact

...

2 Document Indexing

Multi-dimensional Index (Sparse)

Event	Location	Date	Damage	...	DocID
Harvey	Texas	2017	125B	...	d12, d87
Irma	Florida	2017	50B	...	d12, d53
Maria	Puerto Rico	2017	40B	...	d31
Sandy	Florida	2022	60B	...	d87
...	...	...	...	...	...

Each document contributes values to multiple dimensions. Only observed (dimension, value) pairs are stored.

3 Joint Retrieval

Query Decomposition

"Which hurricanes caused > 50B damage in the U.S. after 2010?"

Typed Constraints

Event Type Hurricane
Damage > 50B
Location U.S.
Date > 2010

Joint Match Over constraints



4 Generation

Ranking & ID intersection Top-K Evidence



d12



d87



d53

...

...

Final Answer

Hurricanes Irma,
Harvey, Sandy.